

Shared-completion bounds for incomplete-benchmark decision stability

A technical contribution test, not an additional application

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Decision. The investigation produces a substantive extension of the current fixed-matrix analysis: sharp endpoints of winner-stability radii over an admissible family of incomplete rank evidence. The lower endpoint decomposes over rivals; the upper endpoint generally does not. Individually optimal pairwise completions can be incompatible. An exact constructed example and an existing-evidence experiment both exhibit this obstruction.

No manuscript change. CERT-Bench R3 remains frozen. The results below do not replace its 122-task BeyondArena radius or its other benchmark results. This note is a separate research artifact. Its proofs and computations were developed within this project, not independently peer reviewed.

Main findings

An exact three-task, three-model construction has 125 compatible weak-order completions. Its sharp upper winner radius is $1/5$, while separately optimized pairwise radii suggest $2/9$. Allowing new ties changes the sharp endpoint: it would be $1/6$ if those ties were excluded.

The released-evidence experiment keeps all 122 complete 11-model rank rows and the nine-model relative orders on 20 additional BeyondArena tasks. On this *declared coarsened ordinal evidence*, the following endpoints are sharp:

Fixed candidate	Lower endpoint	Sharp upper	Pairwise upper relaxation
TabPFN-2.6	0	$96/1207 = 0.07953604$	$96/1207$
RealMLP	0	$55/2414 = 0.02278376$	$69/2698 = 0.02557450$
Other nine models	0	0	0

For RealMLP, the relaxation overstates the upper endpoint by $64/22933 = 0.00279074$, or $128/1045 = 12.2488\%$ relative to the sharp endpoint. A compatible rank completion attains $55/2414$. An independent integer dynamic program proves that no compatible completion exceeds it; this conclusion does not depend on a floating-point mixed-integer solver’s optimality flag.

Evidence restriction. The 40 unspecified positions in the selected representation are not asserted to be the original raw missing-cell pattern. Exact raw score/failure records were not recovered. Additional observed scores, metric bounds or failure semantics could narrow the completion family. The endpoint matrices are hypothetical proof witnesses, not new evaluations or imputed estimates. These are sharp endpoints for the declared ordinal family, not claimed sharp endpoints for the unrecovered full raw record.

Novelty restriction. Necessary/possible winners, incomplete-rank aggregation, TV optimization and ILP are established foundations [1, 2, 3, 5]. The candidate contribution is their specific quantitative extension to sharp, rank-consistent task-weight stability envelopes, with a verified shared-completion obstruction. A first-in-literature claim is not established by the present search.

1 Fixed scope, incomplete evidence, and sharp endpoints

There are D tasks and M models. For each task d , let the nonempty finite set $\mathcal{P}_d \subset [0, 1]^M$ contain every admissible full normalized rank-loss row. Ranks are lower-is-better, use average positions for ties, and are normalized by $M - 1$. Observed order and equality relations are preserved. Unspecified models are inserted into a full weak order, not assigned independent rank intervals. A shared completion is $Z \in \mathcal{C} = \prod_{d=1}^D \mathcal{P}_d$.

Task scope, candidate scope, and nominal weights $w^0 \in \Delta_D$ are held fixed throughout, where $\Delta_D = \{w \in \mathbb{R}^D : w \geq 0, \sum_d w_d = 1\}$. For gap vector g , define the established TV boundary problem

$$\Phi(g) = \inf \left\{ \frac{1}{2} \|w - w^0\|_1 : w \in \Delta_D, w^\top g \leq 0 \right\}. \quad (1)$$

Its value is $+\infty$ if the feasible set is empty and 0 if the nominal gap is already nonpositive. For a fixed candidate i ,

$$\rho_i(Z) = \min_{j \neq i} \Phi(g^{ij}(Z)), \quad g_d^{ij}(Z) = Z_{dj} - Z_{di}. \quad (2)$$

The sharp completion endpoints are

$$\underline{\rho}_i = \min_{Z \in \mathcal{C}} \rho_i(Z), \quad \bar{\rho}_i = \max_{Z \in \mathcal{C}} \rho_i(Z). \quad (3)$$

The finite completion family ensures attainment, allowing $+\infty$. The bracket $[\underline{\rho}_i, \bar{\rho}_i]$ is an envelope: it need not equal the set of attainable radii. It is neither a confidence interval nor a probability model for missing scores.

Proposition 1 (Sharp lower endpoint and pairwise upper relaxation). *For $j \neq i$, define $\ell_d^j = \min_{p \in \mathcal{P}_d} (p_j - p_i)$ and $u_d^j = \max_{p \in \mathcal{P}_d} (p_j - p_i)$. Then*

$$\underline{\rho}_i = \min_{j \neq i} \Phi(\ell^j), \quad \bar{\rho}_i \leq U_i^{\text{pair}} := \min_{j \neq i} \Phi(u^j). \quad (4)$$

Each $\Phi(u^j)$ is the sharp upper pairwise endpoint. Equality in the second statement is certified by a single admissible completion with radius U_i^{pair} ; separate pairwise witnesses do not suffice.

Proof. If $g \leq h$ componentwise, nonnegative task weights give $\{w : w^\top h \leq 0\} \subseteq \{w : w^\top g \leq 0\}$; hence $\Phi(g) \leq \Phi(h)$. Row independence makes each gap vector ℓ^j and u^j attainable for its particular rival. Interchanging the two finite minima gives $\min_Z \min_j \Phi(g^{ij}(Z)) = \min_j \Phi(\ell^j)$. Conversely, $\max_Z \min_j \Phi(g^{ij}(Z)) \leq \min_j \max_Z \Phi(g^{ij}(Z)) = \min_j \Phi(u^j)$. The last maxima can use different completions. A shared witness attaining the upper bound proves equality. $\square$

Corollary 1 (Evidence tightening). *If consistent additional evidence gives a nonempty family $\mathcal{F} \subseteq \mathcal{C}$, then $\underline{\rho}_i(\mathcal{C}) \leq \underline{\rho}_i(\mathcal{F}) \leq \bar{\rho}_i(\mathcal{F}) \leq \bar{\rho}_i(\mathcal{C})$. Thus a coarse-family upper bound remains an outer bound after refinement, but its endpoint witness may cease to be admissible. This follows by set inclusion.*

2 A strict shared-completion obstruction, including ties

Consider three equally weighted tasks and models A, B, C . The observed loss rows are

Task	A	B	C
1	2	0	unspecified
2	1	unspecified	0
3	1	unspecified	2

Only the shown order relations are imposed. The unspecified value can be below, between, above, or tied with either of the two distinct observed values. There are five weak orders per row and $5^3 = 125$ shared completions.

In task 1, placing C above both observed losses, so $B < A < C$, maximizes both rival-minus- A gaps componentwise. In task 2, placing B above both observed losses, so $C < A < B$, does the same. These dominance replacements cannot decrease A 's stability under any nonnegative weights. On task 3, only three nondominated weak orders remain:

Task 3 order	(g_3^B, g_3^C)	ρ_A	Role
$A < B < C$	$(1/2, 1)$	$1/6$	Best for gap to C
$A < (B = C)$	$(3/4, 3/4)$	$1/5$	Joint optimum
$A < C < B$	$(1, 1/2)$	$1/6$	Best for gap to B

For the joint optimum the gap vectors are

$$g^B = (-1/2, 1/2, 3/4), \quad g^C = (1/2, -1/2, 3/4). \quad (5)$$

Each nominal gap is $1/4$. Moving weight from the largest gap $3/4$ to the smallest gap $-1/2$ reduces the aggregate gap by $5/4$ per unit moved. The required TV movement is $(1/4)/(5/4) = 1/5$, below the donor's available mass $1/3$. This proves $\bar{\rho}_A = 1/5$ after the dominance reductions.

In contrast, the coordinatewise maximal pairwise vectors are $u^B = (-1/2, 1/2, 1)$ and $u^C = (1/2, -1/2, 1)$. Their boundary radii are $(1/3)/(3/2) = 2/9$. The required last positions of B and C on task 3 conflict: no single row can have both gaps equal to 1. Therefore

$$\bar{\rho}_A = \frac{1}{5} < \frac{2}{9} = U_A^{\text{pair}}. \quad (6)$$

Direct rational enumeration gives exactly the attainable set $\{0, 1/15, 1/6, 1/5\}$, independently checked with fresh primal LPs. In particular, the envelope $[0, 1/5]$ does not imply all intermediate radii are attainable. Disallowing new ties removes the middle joint-optimum row and gives upper endpoint $1/6$.

Interpretation. The obstruction is not that a challenger reaches a nominal winner before becoming globally feasible, as in R3. Here the issue is that the strongest evidence completion against one rival is inconsistent with that against another. The completion is required to be shared across every rival and every admissible reweighting.

3 An exact fixed-budget formulation

Let $B_\epsilon = \{w \in \Delta_D : \text{TV}(w, w^0) \leq \epsilon\}$ and define

$$\eta_i^*(\epsilon) = \max_{Z \in \mathcal{C}} \min_{j \neq i} \min_{w \in B_\epsilon} w^\top g^{ij}(Z). \quad (7)$$

Proposition 2 (One completion, all rivals, all permitted weights). *$\eta_i^*(\epsilon) > 0$ if and only if some compatible completion has $\rho_i(Z) > \epsilon$. For explicitly enumerated row patterns, the value in (7) is the optimum of a mixed-integer linear program.*

Proof. The TV ball is compact. Absence of a nonpositive pairwise margin on that ball is therefore equivalent to a strictly positive minimum margin against every rival. A finite maximum over completions preserves the existence statement. Weak nonnegativity is insufficient at a tie.

For fixed g , the dual of its TV-ball support minimization is

$$\max_{\nu, t, v} \nu + (w^0)^\top v - 2\epsilon t \quad \text{s.t.} \quad \nu + v_d \leq g_d, \quad -t \leq v_d \leq t, \quad t \geq 0. \quad (8)$$

This follows by introducing absolute-deviation variables in the primal and using LP duality. Both primal feasibility and finite optimality hold. Index row patterns by p . Introduce $y_{dp} \in \{0, 1\}$ with $\sum_p y_{dp} = 1$. For each rival j , use its own (ν_j, t_j, v^j) , with the same y across all rivals. Maximize η subject to

$$\nu_j + v_d^j \leq \sum_{p \in \mathcal{P}_d} y_{dp}(p_j - p_i), \quad (9)$$

$$-t_j \leq v_d^j \leq t_j, \quad t_j \geq 0, \quad (10)$$

$$\eta \leq \nu_j + \sum_d w_d^0 v_d^j - 2\epsilon t_j \quad (j \neq i). \quad (11)$$

Strong duality establishes equality with (7). There are no products between continuous weight variables and binary completion variables. Relaxing y to mixtures is only a relaxation: a mixture need not be a full rank row. $\square$

The program is exact as a mathematical formulation. Floating-point MILP output still requires tolerance checks and does not itself prove an exact endpoint. No general polynomial-time guarantee is claimed; possible-winner problems and practical ILP precedents already exist for partial rankings and partial chains [3, 4].

Bounded implementation. Since $g_d \in [-1, 1]$, a dual optimum can be chosen with $t \leq 1$, $|\nu| \leq 1$ and $|v_d| \leq 1$. To see this, put $m = \min_d g_d$. For a cutoff $c \in [m, \max_d g_d]$, take $t = (c - m)/2$, $\nu = (c + m)/2$, and $v_d = \min(g_d, c) - \nu$. These realize

$$\min_{w \in B_\epsilon} w^\top g = \max_{c \in [m, \max_d g]} \left\{ \sum_d w_d^0 \min(g_d, c) - \epsilon(c - m) \right\}. \quad (12)$$

For any feasible dual, setting $c = \nu + t$ shows its objective is bounded above by this expression, or by the endpoint $c = m$ if $c < m$; cutoffs beyond $\max g$ do not improve it. This also proves the stated bounds. TV extrema are an established foundation, not a separate novelty claim [5].

4 Exact upper certification without trusting a solver flag

For the uniform 142-task experiment, all full normalized average ranks are on the grid $\{0, 1/20, \dots, 1\}$. Scale a rival gap by 20 to get integers a_d , and let $m = \min_d a_d$. Equation (12) gives the exact identity

$$\min_{w \in B_\epsilon} \sum_d w_d \frac{a_d}{20} = \frac{1}{20D} \max_{v \in \{m, \dots, \max a\}} \left\{ \sum_d \min(a_d, v) - D\epsilon(v - m) \right\}. \quad (13)$$

The expression is piecewise linear between integer gaps, so a maximizing cutoff can be chosen on the integer grid. For $D\epsilon = k + h$, k integer and $0 < h < 1$, the $(k+1)$ st largest gap is a maximizing cutoff: the slope changes from positive to negative there. Equivalently, remove the top k gaps and a fraction h of the next gap, moving that weight to a minimum-gap row.

The RealMLP certificate

Set $\epsilon = 55/2414$ so that $D\epsilon = 55/17 = 3 + 4/17$. Use two rivals, $A = \text{TabICLv2}$ and $B = \text{TabPFN-2.6}$. Across the 122 complete rows, their integer gap sums are -126 and -208 ; their minima are $m_A = -16$ and $m_B = -18$. Every admissible pattern on the 20 partial rows has gaps no smaller than these minima. Thus these minima are fixed across all completions.

A maximizing cutoff is the fourth largest full gap. Known complete rows imply $v_A \in \{18, 19, 20\}$ and $v_B \in \{14, 15, \dots, 20\}$. Only 21 cutoff pairs need checking, while retaining *every* compatible joint gap pair on each partial row. Let $K_A(v_A)$ and $K_B(v_B)$ be the corresponding sums of clipped known gaps. For a partial row r , let Q_r be its possible pairs (a, b) of integer gaps to the two rivals.

For a fixed cutoff pair, define $F_r(s)$ as the largest accumulated clipped B -gap over the first r partial rows with accumulated clipped A -gap equal to s . Initialize $F_0(0) = 0$ and other states $-\infty$. The exact recurrence is

$$F_r(s) = \max_{(a,b) \in Q_r} \{F_{r-1}(s - \min(a, v_A)) + \min(b, v_B)\}. \quad (14)$$

For the first rival to have a strictly positive support margin,

$$s > \frac{55}{17}(v_A - m_A) - K_A(v_A). \quad (15)$$

Since s is integer, the lower state threshold is the floor of the right-hand side plus 1. For all such states, the second rival's scaled support residual is at most

$$K_B(v_B) + \max_{s \text{ above threshold}} F_{20}(s) - \frac{55}{17}(v_B - m_B). \quad (16)$$

Exact integer/rational evaluation across all 21 cutoff pairs gives maximum $-33/17$, strictly negative (normalized margin $-33/48280$). No completion can make both rival margins positive at this radius. Hence $\bar{\rho}_{\text{RealMLP}} \leq 55/2414$. The stored shared-completion witness has exact minimum pairwise radius $55/2414$, proving equality. The certificate uses necessary conditions from two rivals to exclude *all* completions; it is not an enumeration of 381^{20} complete matrices.

5 Existing-evidence experiment and its limits

The input files are the byte-identical author-derived matrices `primary11__all__rank__equal__normalized_matrix.csv` and `coverage9__all__rank__equal__normalized_matrix.csv` from the R3 supplementary package. Their dimensions are 122×11 and 142×9 . Restricting the complete matrix to the nine common models and recomputing ranks reproduces the bridge’s 122 common rows with zero discrepancy.

On the 20 remaining tasks, only the nine-model relative order is retained; TabDPT and TabPFN-2.6 positions are unspecified. Inserting the two models, including possible ties, gives 381 full rank rows per task. The completion family therefore has $381^{20} \approx 4.1543 \times 10^{51}$ matrices and preserves 7,430 observed unordered-pair relations. No distribution over those matrices is used.

TabPFN-2.6. The sharp lower endpoint is zero: a valid lower witness makes RealMLP the unique nominal winner. A single upper witness attains the pairwise upper relaxation 96/1207. Proposition 1 therefore certifies the sharp envelope $[0, 96/1207]$.

RealMLP. Its sharp lower endpoint is also zero. Separate pairwise upper optimization gives 69/2698, but no shared completion attains that radius. The shared endpoint witness and Section 4’s exact obstruction give $[0, 55/2414]$. Its endpoint witness has the following exact boundary radii:

Rival	Radius	Rival	Radius
LinearModel	319/923	XGBoost	43/284
RandomForest	79/284	TabM	137/1420
ExtraTrees	963/4544	TabDPT	669/4544
CatBoost	301/3408	TabPFN-2.6	31/1349
LightGBM	91/710	TabICLv2	55/2414

For each of the other nine models, a zero pairwise upper relaxation proves that it cannot be a unique nominal winner under any completion. This does not rule out co-winner status by an independent argument and is not presented as such.

What this does not show. The original complete-case result $41/2196 \approx 0.01867031$ for TabPFN-2.6 uses 122 equal task weights. The new endpoints use 142 equal task weights and a different evidence family. They do not correct, contradict, or improve that original number. No result is a probability of reversal, a deployment recommendation, or a universal robustness scale. No new result is asserted for top- k or global feasible-winner entry.

The inherited retrieval manifest records original raw hashes and byte counts, but not exact download URLs. Neither recovered archive supplied the original raw score/failure mask. G1 thus deliberately coarsens the released evidence. It does not assert that all 40 unspecified positions were originally missing, that unsupported evaluations have ordinary latent scores, or that every ordinal completion is compatible with unrecovered metric bounds. Those distinctions must be resolved before a sharp *original-raw* missingness claim.

6 Validation, contribution boundary, and next decision

The calculations were executed with Python 3.13.5, NumPy 2.3.5, pandas 2.2.3 and SciPy 1.17.0. Exact endpoint and upper-obstruction checks use rational/integer arithmetic. Independent LP/MILP comparisons use floating-point HiGHS solves.

Validation	Cases	Maximum discrepancy
Insertion generator vs independent weak-order enumeration	68	0
Rational pairwise radii vs fresh primal LPs	400	2.78×10^{-16}
Constructed-example full completion enumeration	125	4.16×10^{-17}
Sharp lower / relaxed upper vs exhaustive candidate profiles	72	0 violations
Joint MILP vs full completion enumeration + primal LP	30	1.11×10^{-16}
Integer clipping identity vs fresh primal support LP	200	1.07×10^{-14}
Real endpoint witness pairwise checks	20	3.89×10^{-16}
RealMLP exact upper-obstruction cutoff pairs	21	Exact integer PASS

Both real endpoint witnesses passed all 284 full-row membership checks and preserved all 7,430 observed relations per completion. The mixed-integer solver also returned zero joint margin at 55/2414 with zero reported MIP gap, but this numerical result is supplementary to the integer upper certificate. Code-path independence does not eliminate correlated specification errors; external mathematical review remains appropriate.

What can credibly be claimed

The exact object differs from obtaining a point ranking by completing missing scores [1] or estimating identifiable scores from incomplete samples [6]. It quantitatively refines possible/necessary unique-winner analysis [2, 3] by retaining a continuum of TV task-weight perturbations. Its mathematical foundations remain familiar: minimax ordering, TV support duality, partial rankings, and discrete optimization [5, 4]. The present search does not establish priority for the complete combination or an absence of equivalent earlier formulations.

The contribution candidate is therefore specific: *sharp completion endpoints for task-reweighting stability, a proof that sharp pairwise endpoints can overstate joint winner stability, and checkable shared-completion endpoint certificates*. The real-data relaxation gap demonstrates that respecting one shared completion changes the answer, rather than merely adding computation.

Next decision

G1 is a mathematical/computational pass for its declared ordinal model. Keep R3 as the active manuscript. Before integration, complete the closest-prior-work comparison and choose whether to report the precisely labeled ordinal-evidence experiment or tighten it using the original raw mask and metric/failure constraints. The latter requires exact artifact recovery, not a different benchmark or a newer living-registry snapshot. Added consistent constraints must narrow the envelope; endpoint values must not be protected by omitting newly recovered evidence. Integration should replace repetitive exposition, not append another application or inflate dataset counts.

References

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Reproduction and source provenance

Run `python code/reproduce.py` from the package root. It verifies input hashes, rebuilds pattern sets, calculates endpoint bounds, recomputes the exact upper obstruction, executes independent tests, and produces the final result summary. The stored RealMLP endpoint is verified as a proof witness; the optional `-rediscover` flag also reruns the mixed-integer search. Alternative optimal solver witnesses need not be byte-identical.

The public technical literature above was checked on 22 September 2026. The primary empirical inputs are the two derived rank matrices in the author-supplied R3 supplement, not new live downloads. Source hashes and the original raw-artifact retrieval manifest are included. No third-party raw parquet, author biographies, credentials, manuscript author-role claims, or repository DOI is included.

The G1 validation, proofs and output tables are separated from the journal-facing manuscript. The R3 main PDF matches its original package member by SHA-256:

e6d707930466ad3cfd68c56090a2cdd5e9b42098781f9d98be27245da3486857

The complete source register records what was checked, which foundations are inherited, and the limit of the novelty search. The note does not claim an independent human validation, a formal proof-assistant verification, or journal acceptance readiness for the extension.